Roberto Bloom

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SUMMARY

I build the software that turns drone imagery into georeferenced 3D. My pipeline runs COLMAP Structure-from-Motion, fits solved camera positions to EXIF GPS in a local ENU frame, trains a Gaussian splat on the result, and delivers 3D Tiles that render in CesiumJS at true latitude, longitude, and elevation. Recent captures came in at 0.24 m 3D RMSE and 0.107 m horizontal residual across 119 of 120 cameras. I also hold a Part 107 and have flown over 100 commercial missions, so I plan the capture around what the processing actually needs.

TECHNICAL SKILLS

3D reconstruction: Gaussian Splatting, COLMAP and pycolmap, Nerfstudio, photogrammetry, LiDAR point clouds, SPZ, 3D Tiles, RealityCapture, CloudCompare, DJI Terra

Geospatial and geodesy: CesiumJS, Three.js, QGIS, ArcGIS Pro, WGS84 / ECEF / ENU, Sim3 alignment, geoid and terrain correction, EPSG and CRS handling, PDAL, COPC, GeoJSON

Software: Python, TypeScript, React Native and Expo, Next.js, Flask, FastAPI, PHP, Docker, AWS S3, Git

UAS and sensors: FAA Part 107, LAANC, DJI Matrice and M3M, RTK mission planning, nadir and oblique capture, GSD targeting, KML/KMZ, multispectral, thermal, RGB

EXPERIENCE

GIS Developer, GeoInformatica Consulting *contract, remote* Jun 2026 – Aug 2026

- Built the pipeline that turns raw drone photos into georeferenced 3D models, covering COLMAP SfM, EXIF GPS to local ENU alignment, Nerfstudio splat training, SPZ-compressed 3D Tiles, and CesiumJS display at true world coordinates.
- **Hit 0.24 m 3D RMSE and 0.107 m horizontal residual** (119 of 120 solved cameras) and surfaced both in-app as a per-reconstruction accuracy readout.
- Tracked down why models were sitting tens of metres above the terrain. GPS altitude is height above the WGS84 ellipsoid, not above sea level, and the horizontal fit looks perfect the whole time it is wrong.
- Wrote a supervised classifier for Gaussian scenes and ran it across a 1.3M-Gaussian capture. Operator QA taught it classes the unsupervised clustering never produced, including one for dead vegetation. Results export to 3D Tiles and GeoJSON for ArcGIS.
- Shipped the in-app 3D viewer so field crews open GPS-located models from their own imagery, plus ingestion: EXIF extraction, map pinning, mission grouping, offline-first storage.

Reality Capture Engineer and Lead Drone Operator, Firescore AI *remote, with California field ops* Feb 2025 – May 2026

- Flew and planned over 100 UAV missions for wildfire inspection, vegetation analysis, parcel review, and 3D reconstruction, across RGB, multispectral, thermal, and LiDAR-supported workflows.
- Wrote the capture standards the team flew to: obstacle clearance, height tiers, image overlap, GSD targets, RTK setup, lighting consistency. Reflights dropped because the parameters were decided before anyone left the truck.
- Built the first working splat-to-CesiumJS path, using COLMAP, EXIF GPS to ENU alignment, parcel overlays, nearest-source-image lookup, and RMSE checks on the georeferencing.
- Developed and tested a Windows SDK capture workflow for the Sony IMX454 multispectral sensor, covering integration, capture modes, and reviewing processed output for vegetation work.
- Handled large datasets across DJI Terra, RealityCapture, CloudCompare, QGIS, ArcGIS Pro, and AWS S3, working alongside fire scientists, GIS staff, and engineers.

EARLIER

Inorganic Lab Technician, ALS Global. Sample prep for TCLP analysis under SOP and documentation controls; eLIMS/GEL data capture. May – Oct 2024

Undergraduate Research Assistant, Trevor Nichols Research Center. Insecticide trials on fruit crop pests; microscopy work quantifying spotted wing drosophila. Summers 2022, 2023

EDUCATION AND CERTIFICATION

B.S. Entomology, Michigan State University Apr 2024

FAA Part 107 Remote Pilot Certificate. Valid driver's license.